Direct Smoothness-Constrained Spectral Reconstruction for Penalized Least-Squares Trends: Exact Equivalence, Endpoint Geometry, and Nonzero Difference Means

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ABSTRACT

Finite-difference penalized least-squares (PLS) trends may be described by the penalty $\lambda \in [0, \infty]$ or by a normalized trace smoothness $s \in [0, 1]$. We develop and audit a direct spectral-weight representation in which the numerical reconstruction takes s as its input and does not explicitly form λ. A single strictly monotone scalar equation determines the coupled shrinkage weights. We prove rank and nullspace properties of the difference operator, an exact characterization of the PLS spectral-weight family, existence and uniqueness of the constrained weight solution, equivalence with the conventional penalty-based estimator, endpoint limits, sensitivity bounds, and formulas for a fixed nonzero mean of differences. For the observed-difference plug-in mean, we derive the full data-dependent smoother matrix and its effective-degrees-of-freedom correction, which differs from the customary conditional trace. We give a bracketed log-odds numerical algorithm, complexity accounting and falsifiable benchmark protocols. The scalar root is algebraically equivalent to inversion in log-penalty coordinates: this is neither a new regularization family nor an inversion-free algorithm. No numerical superiority is claimed without a matched accuracy–time study.

1. Introduction and contribution boundary

Quadratic difference-penalty smoothing is a classical model, originating in graduation and time-series filtering (Whittaker, 1923). Its PLS formulation permits nonzero reference means of differences (Guerrero, 2007). Guerrero's controlled-smoothness framework already allows users to prescribe a percentage of trend smoothness (Guerrero, 2008), and the relation to criterion-based smoothing selection has been studied (Cortés-Toto, Guerrero and Reyes, 2017). Prescribing an effective dimension in place of a raw regularization parameter is also familiar in smoothing-spline software, including `smooth.spline` (rsm, 2026). Thus neither user-controlled smoothness nor spectral shrinkage is new.

The question addressed here is more specific: *can the established finite-difference PLS estimator be reconstructed from a bounded smoothness target using coupled spectral weights, without explicitly forming the penalty coefficient in the fitting path?* This question has an affirmative algebraic answer. Whether the resulting implementation is faster or more robust than an equivalently engineered log-penalty implementation is an empirical hypothesis rather than a theorem.

The analysis provides (i) a characterization of the original penalty family by its spectral weights; (ii) a globally bracketed scalar equation in reference spectral log-odds; (iii) exact reconstruction for any integer $1 \leq d < n$ and fixed, possibly nonzero, difference mean μ; (iv) endpoint and sensitivity results; and (v) an exact effective-degrees-of-freedom correction when the difference mean is estimated as $\widehat{\mu}(y) = \mathrm{mean}(D_d y)$. Mathematical statements are in exact arithmetic unless numerical evaluation is explicitly discussed.

This is neither a proposed new statistical regularization family nor a closed-form inverse of smoothness. Efficient structured fitting is established prior art (Weinert, 2007); a modern Whittaker–Henderson treatment addresses related statistical and likelihood questions (Biessy, 2026). A matched validation and runtime study is required before a computational advantage can be claimed.

Notation and assumptions. $n \geq 2$ is the number of observations; d is an integer with $1 \leq d < n$; and $r = n - d \geq 1$. All vectors and matrices are real. Define the unscaled dth forward-difference operator $D_d \in \mathbb{R}^{r \times n}$ by

$$(D_d \tau)_i = \sum_{k=0}^{d} (-1)^{d-k} \binom{d}{k} \tau_{i+k}, \qquad i = 1, \ldots, r. \quad (1)$$

$\mathbf{1}_r$ is the r-vector of ones. Unless specified otherwise, $\mu \in \mathbb{R}$ is a *fixed* reference mean; conditional effective degrees of freedom refers to treating it as fixed even if its numerical value was computed from data. For a matrix A, $\mathcal{N}(A)$ is its nullspace and $\mathrm{tr}(A)$ its trace. The operators $\odot$ and $\oslash$ mean componentwise multiplication and division. Exact endpoints refer to the limiting fitted trends rather than evaluation of a floating-point matrix at $\lambda = \infty$.

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2. Difference-penalty geometry

Lemma 2.1 (Rank, nullspace and affine feasibility)**.** *For every integer $1 \le d < n$, D_d has row rank $r = n-d$. Its nullspace consists precisely of sampled discrete polynomials of degree at most $d-1$ and has dimension d. For each fixed $\mu \in \mathbb{R}$, the affine constraint $D_d\tau = \mu \mathbf{1}_r$ is nonempty. In particular,*

$$(p_\mu)_i = \mu \binom{i-1}{d}, \qquad i = 1, \dots, n,$$

is a feasible vector.

Proof. By (1), the first r columns of D_d form an upper triangular $r \times r$ matrix with diagonal entries $(-1)^d$, which are nonzero. Hence $\operatorname{rank}(D_d) = r$ and rank–nullity gives $\dim \mathcal{N}(D_d) = n - r = d$. The d sampled polynomials $1, i, \dots, i^{d-1}$ are linearly independent for $n \ge d$ (their first d rows form a Vandermonde matrix), and their dth differences vanish. Thus they form a basis for $\mathcal{N}(D_d)$. Finally, Pascal's identity $\binom{i}{k} - \binom{i-1}{k} = \binom{i-1}{k-1}$ applied d times gives $D_d(\binom{i-1}{d})_{i=1}^n = \mathbf{1}_r$. Multiplying by μ proves feasibility. □

For $y \in \mathbb{R}^n$, $\lambda \ge 0$ and fixed μ, consider the classical quadratic objective

$$J_{\lambda,\mu}(\tau) = \|y - \tau\|_2^2 + \lambda \|D_d\tau - \mu \mathbf{1}_r\|_2^2 . \tag{2}$$

Let $Q_d = D_d^\top D_d$ and $H_\lambda = (I_n + \lambda Q_d)^{-1}$.

Theorem 2.2 (Unique PLS trend and affine smoother)**.** *For each finite $\lambda \ge 0$, (2) has the unique minimizer*

$$\widehat{\tau}_{\lambda,\mu} = (I_n + \lambda Q_d)^{-1}(y + \lambda D_d^\top \mu \mathbf{1}_r). \tag{3}$$

For any feasible p_μ from Lemma 2.1,

$$\widehat{\tau}_{\lambda,\mu} = p_\mu + H_\lambda(y - p_\mu).$$

Consequently, for fixed μ, the Jacobian with respect to y is H_λ.

Proof. The Hessian of $J_{\lambda,\mu}$ equals $2(I_n + \lambda Q_d)$, which is positive definite because, for every nonzero v, $v^\top(I_n + \lambda Q_d)v = \|v\|_2^2 + \lambda \|D_d v\|_2^2 > 0$. The vanishing first derivative gives (3). Since $D_d p_\mu = \mu \mathbf{1}_r$, $(I_n + \lambda Q_d)p_\mu = p_\mu + \lambda D_d^\top \mu \mathbf{1}_r$. Subtracting this identity from the normal equation gives $(I_n + \lambda Q_d)(\widehat{\tau} - p_\mu) = y - p_\mu$, establishing the affine formula. Differentiating in y with p_μ fixed yields H_λ. □

Take the reduced SVD

$$D_d = U \Sigma V^\top, \quad U^\top U = I_r, \quad V^\top V = I_r, \quad \Sigma = \operatorname{diag}(\sigma_1, \dots, \sigma_r), \tag{4}$$

with $\sigma_j > 0$ and $\delta_j = \sigma_j^2$. The positive eigenvalues of Q_d are precisely the δ_j; its d remaining eigenvalues vanish. Define $w_j(\lambda) = \lambda\delta_j/(1 + \lambda\delta_j)$.

Theorem 2.3 (Spectral filter and normalized trace smoothness)**.** *For each finite $\lambda \ge 0$,*

$$H_\lambda = I_n - V \operatorname{diag}(w_1(\lambda), \dots, w_r(\lambda)) V^\top, \tag{5}$$

$$\nu_{\mathrm{cond}}(\lambda) = \operatorname{tr} H_\lambda = n - \sum_{j=1}^r w_j(\lambda). \tag{6}$$

The normalized conditional smoothness index

$$S_d(\lambda) = \frac{n - \nu_{\mathrm{cond}}(\lambda)}{r} = \frac{1}{r}\sum_{j=1}^r w_j(\lambda) \tag{7}$$

extends continuously and strictly increasingly from $S_d(0) = 0$ to $S_d(\infty) = 1$. For $0 < s < 1$ there is a unique finite $\lambda = \Lambda_d(s) > 0$.

Proof. From (4), $Q_d = V\Sigma^2 V^\top$. On $\mathcal{R}(V)$ the inverse H_λ multiplies the jth singular direction by $(1 + \lambda\delta_j)^{-1} = 1 - w_j$, and on $\mathcal{N}(D_d) = \mathcal{R}(V)^\perp$ it acts as the identity. This proves (5). Taking traces and using $V^\top V = I_r$ gives (6) and (7). Each $\delta_j > 0$, so

$$S_d'(\lambda) = \frac{1}{r}\sum_{j=1}^r \frac{\delta_j}{(1 + \lambda\delta_j)^2} > 0$$

for finite $\lambda \ge 0$. Each summand of (7) tends to zero as $\lambda \downarrow 0$ and to one as $\lambda \uparrow \infty$. Continuity, strict monotonicity, and the intermediate value theorem give the unique inverse in the interior; the endpoints are understood in the extended nonnegative real line. □

Remark 2.4 (Normalization and interpretation)**.** Guerrero's earlier convention with denominator n has upper endpoint r/n for $d > 0$ (Guerrero, 2008). Dividing the same conditional trace reduction by $r = n - d$ produces $S \in [0, 1]$ for every admissible order. Equal S across orders means equal *fraction of penalizable trace degrees of freedom removed*; it does not assert equal curvature, equal fitted values, or the same physical units. When μ depends on y, (6) remains a *conditional* trace identity, not the Jacobian trace of the fully data-dependent estimator; see Section 4.

3. A direct smoothness-constrained weight representation

Fix any reference $\delta_* > 0$ and put

$$a_j = \delta_j/\delta_* > 0, \qquad \alpha_j = \log a_j, \qquad \operatorname{expit}(z) = \frac{1}{1 + e^{-z}}.$$

For $u \in (0, 1)$ define the *coupled spectral-weight family*

$$w_j(u) = \frac{a_j u}{1 - u + a_j u}, \quad G_u(u) = \frac{1}{r}\sum_{j=1}^r w_j(u). \tag{8}$$

This is not a free r-dimensional collection of weights: the common scalar u couples all directions.

Theorem 3.1 (Characterization of the PLS spectral family)**.** *For $w \in (0,1)^r$, the following statements are equivalent:*

1. *There exists $\lambda > 0$ such that $w_j = \lambda\delta_j/(1+\lambda\delta_j)$ for every j.*
2. *All numbers $w_j/[\delta_j(1-w_j)]$ are equal.*
3. *There exists $u \in (0,1)$ for which (8) holds.*

The associated λ and u are unique. Arbitrarily setting all $w_j = s$ is a PLS filter for an interior s only if all positive δ_j are equal.

Proof. If (i) holds, direct rearrangement yields $w_j/(1-w_j) = \lambda\delta_j$, proving (ii). If (ii) holds, write the positive common value as λ; solving $w_j = \lambda\delta_j(1-w_j)$ gives (i). Under (i) let $u = \lambda\delta_*/(1+\lambda\delta_*)$. Then $u/(1-u) = \lambda\delta_*$, and

$$\frac{a_j u}{1-u+a_j u} = \frac{\lambda\delta_j}{1+\lambda\delta_j},$$

which is (iii). Conversely, under (iii), the positive number $\lambda = u/[\delta_*(1-u)]$ gives (i). Each inverse formula is injective, proving uniqueness. Finally, if every $w_j = s \in (0,1)$, condition (ii) says $s/[(1-s)\delta_j]$ is independent of j; hence the δ_j must coincide. The converse is immediate. □

Theorem 3.2 (Unique direct-S solution and explicit root bracket)**.** *For every target $0 < s < 1$, there exists a unique $u_s \in (0,1)$ such that $G_u(u_s) = s$. Equivalently, with*

$$x = \operatorname{logit}(u) = \log\left(\frac{u}{1-u}\right), \qquad G(x) = \frac{1}{r}\sum_{j=1}^{r} \operatorname{expit}(x+\alpha_j),$$

the equation $G(x_s) = s$ has one solution on the real line, satisfying

$$\operatorname{logit}(s) - \max_j \alpha_j \;\le x_s \le\; \operatorname{logit}(s) - \min_j \alpha_j. \quad (9)$$

The weights $w_j(s) = \operatorname{expit}(x_s + \alpha_j)$ attain exact mean s. The closed extensions are $w_j(0) = 0$ and $w_j(1) = 1$.

Proof. Each function $u \mapsto w_j(u)$ is continuous on $[0,1]$, takes the values zero and one at the respective endpoints, and has derivative

$$w_j'(u) = \frac{a_j}{(1-u+a_j u)^2} > 0.$$

Thus their average is a continuous strictly increasing bijection from $[0,1]$ to $[0,1]$, proving existence and uniqueness. In log-odds coordinates,

$$G'(x) = \frac{1}{r}\sum_{j=1}^{r} \operatorname{expit}(x+\alpha_j)[1-\operatorname{expit}(x+\alpha_j)] > 0.$$

Set $x_- = \operatorname{logit}(s) - \max_j \alpha_j$. For every j, $x_- + \alpha_j \le \operatorname{logit}(s)$, so $\operatorname{expit}(x_- + \alpha_j) \le s$, hence $G(x_-) \le s$. Similarly, at $x_+ = \operatorname{logit}(s) - \min_j \alpha_j$, every term is at least s, and $G(x_+) \ge s$. Strict monotonicity gives $x_- \le x_s \le x_+$. Substitution gives the asserted weights. The endpoint extensions follow by letting x tend to $-\infty$ and $+\infty$. □

Remark 3.3 (The hidden penalty coordinate)**.** The scalar root is *not* an entirely new parameterization: from the proof of Theorem 3.1,

$$\lambda = \frac{e^{x_s}}{\delta_*}, \qquad x_s = \log\lambda + \log\delta_*. \quad (10)$$

The algorithm need not evaluate the right side of (10) to form the trend, but a one-dimensional implicit inversion remains. Replacing a stable log-λ root solve by x shifts its argument by a constant. A speed or stability advantage therefore cannot follow from the change of root coordinate alone.

Theorem 3.4 (Exact reconstruction for arbitrary fixed difference mean)**.** *For fixed y, μ and $s \in [0,1]$, let $w(s)$ be the weights of Theorem 3.2, including its endpoint extensions. Define $c_\mu = V^\top y - \mu\Sigma^{-1}U^\top \mathbf{1}_r$. Then*

$$\widehat{\tau}_{s,\mu} = y - V\operatorname{diag}(w(s))c_\mu = y - V\left[w(s)\odot(V^\top y) - \mu\,(w(s)\oslash\sigma)\odot(\quad (11)$$

For each interior s, this is exactly *the unique minimizer of (2) at $\lambda = \Lambda_d(s)$. No explicit λ or matrix inverse is required to evaluate (11) once the weights and SVD are available. At $s = 0$, $\widehat{\tau}_{0,\mu} = y$. At $s = 1$,*

$$\widehat{\tau}_{1,\mu} = (I_n - VV^\top)y + \mu V\Sigma^{-1}U^\top\mathbf{1}_r, \quad (12)$$

the unique least-squares projection of y onto $\{\tau : D_d\tau = \mu\mathbf{1}_r\}$.

Proof. Project the normal equation (3) into $\mathcal{R}(V)$ and its orthogonal complement. Write $z = V^\top\widehat{\tau}$ and $z_y = V^\top y$. Since $D_d^\top\mathbf{1}_r = V\Sigma U^\top\mathbf{1}_r$, the normal equation in the jth right singular direction is

$$(1+\lambda\sigma_j^2)z_j = z_{y,j} + \lambda\mu\sigma_j(U^\top\mathbf{1}_r)_j.$$

Solving and rearranging yields

$$z_j = z_{y,j} - \frac{\lambda\sigma_j^2}{1+\lambda\sigma_j^2}\left(z_{y,j} - \frac{\mu}{\sigma_j}(U^\top\mathbf{1}_r)_j\right).$$

The orthogonal complement component is unchanged from y, because D_d vanishes there. Theorems 3.1 and 3.2 identify the coefficient as $w_j(s)$ for $\lambda = \Lambda_d(s)$, proving (11). At $s = 0$ all weights vanish. At $s = 1$ all weights equal one and (12) follows. The latter vector is feasible: multiplying by $D_d = U\Sigma V^\top$ gives $D_d\widehat{\tau}_{1,\mu} = \mu UU^\top\mathbf{1}_r = \mu\mathbf{1}_r$. Every other feasible τ has the form $\widehat{\tau}_{1,\mu} + v$ with $v \in \mathcal{N}(D_d)$; $\widehat{\tau}_{1,\mu} - y = -Vc_\mu$ lies in $\mathcal{N}(D_d)^\perp$. By the Pythagorean identity, adding a nonzero feasible direction increases squared distance to y, proving uniqueness of the projection. □

Corollary 3.5 (One-dimensional penalized spectrum)**.** *If $r = n - d = 1$, the unique weight is $w_1(s) = s$ for all $s \in [0,1]$. No scalar root solve is needed: $\widehat{\tau}_{s,\mu} = y - sVc_\mu$.*

Proof. The average of a single weight equals the weight itself. Apply (11). □

4. Sensitivity and estimated difference means

Theorem 4.1 (Smoothness sensitivity and conditioning)**.** *For each interior target $s \in (0,1)$, let $b_j(s) = w_j(s)[1 - w_j(s)]$ and $B(s) = r^{-1}\sum_{j=1}^{r} b_j(s) > 0$. Then the spectral weights and the fixed-μ fitted trend are differentiable and satisfy*

$$\frac{\mathrm{d}w_j}{\mathrm{d}s} = \frac{b_j(s)}{B(s)}, \quad 0 < \frac{\mathrm{d}w_j}{\mathrm{d}s} \le r, \qquad \sum_{j=1}^{r} \frac{\mathrm{d}w_j}{\mathrm{d}s} = r, \tag{13}$$

$$\frac{\partial \widehat{\tau}_{s,\mu}}{\partial s} = -V \operatorname{diag}\left(\frac{b_1(s)}{B(s)}, \ldots, \frac{b_r(s)}{B(s)}\right) c_\mu. \tag{14}$$

For all $s, t \in [0,1]$,

$$\|\widehat{\tau}_{s,\mu} - \widehat{\tau}_{t,\mu}\|_2 \le r|s-t|\,\|c_\mu\|_2\,. \tag{15}$$

For fixed s and two numeric means μ_1, μ_2,

$$\|\widehat{\tau}_{s,\mu_1} - \widehat{\tau}_{s,\mu_2}\|_2 = |\mu_1 - \mu_2| \left[\sum_{j=1}^{r} \frac{w_j(s)^2 (U^\top \mathbf{1}_r)_j^2}{\sigma_j^2}\right]^{1/2}. \tag{16}$$

Moreover the derivative of the scalar spectral map obeys

$$G'(x_s) = s(1-s) - \frac{1}{r}\sum_{j=1}^{r}[w_j(s) - s]^2 \le s(1-s), \tag{17}$$

and hence $\mathrm{d}x_s/\mathrm{d}s \ge [s(1-s)]^{-1}$.

Proof. Differentiate $G(x_s) = s$ and use $G'(x_s) = B(s) > 0$ to obtain $\mathrm{d}x_s/\mathrm{d}s = 1/B(s)$. The chain rule then gives $\mathrm{d}w_j/\mathrm{d}s = b_j(s)/B(s)$. Because $rB(s) = \sum_k b_k(s) \ge b_j(s)$, the derivative is at most r; summing the derivatives gives r. Differentiate (11) with y, μ fixed for (14). The mean-value theorem and (13), extended to endpoint limits by continuity, yield $|w_j(s) - w_j(t)| \le r|s-t|$. Since $V^\top V = I_r$,

$$\|V \operatorname{diag}(w(s) - w(t))c_\mu\|_2 \le \max_j |w_j(s) - w_j(t)|\,\|c_\mu\|_2\,,$$

which proves (15). The μ-dependence of (11) is affine:

$$\widehat{\tau}_{s,\mu_1} - \widehat{\tau}_{s,\mu_2} = (\mu_1 - \mu_2) V \operatorname{diag}(w(s))\Sigma^{-1}U^\top \mathbf{1}_r.$$

Its Euclidean norm equals the right side of (16) because V is an isometry on $\mathbb{R}^r$. Finally, using $s = r^{-1}\sum_j w_j$,

$$G'(x_s) = \frac{1}{r}\sum_j (w_j - w_j^2) = s - \left(s^2 + \frac{1}{r}\sum_j (w_j - s)^2\right),$$

which gives (17). Inverting the strictly positive derivative establishes the lower bound on $\mathrm{d}x_s/\mathrm{d}s$. □

Remark 4.2 (Conditioning is not eliminated)**.** The factor $\mathrm{d}x_s/\mathrm{d}s$ becomes large toward both endpoints. This is also the conditioning of a log-penalty inverse, up to the constant shift in (10). In contrast, (13) bounds the sensitivity of each *weight* to the requested smoothness. A well-conditioned weight curve does not imply that its scalar log-odds root is well-conditioned, or that an inaccurate computed SVD can be repaired by reparameterization.

Proposition 4.3 (Endpoint penalty scales)**.** *With n, d and the positive spectrum fixed, define $A = r^{-1}\sum_{j=1}^{r} \delta_j > 0$ and $B = r^{-1}\sum_{j=1}^{r} \delta_j^{-1} > 0$. Then*

$$S_d(\lambda) = A\lambda + O(\lambda^2) \qquad (\lambda \downarrow 0), \tag{18}$$

$$1 - S_d(\lambda) = B/\lambda + O(\lambda^{-2}) \qquad (\lambda \uparrow \infty). \tag{19}$$

Consequently $\Lambda_d(s) = s/A + O(s^2)$ as $s \downarrow 0$ and $\Lambda_d(s) = B/(1-s) + O(1)$ as $s \uparrow 1$.

Proof. For each fixed $\delta_j > 0$, $\lambda\delta_j/(1 + \lambda\delta_j) = \lambda\delta_j + O(\lambda^2)$ as $\lambda \to 0$. The finite average gives the first expansion. Similarly $1/(1 + \lambda\delta_j) = (\lambda\delta_j)^{-1} + O(\lambda^{-2})$ as $\lambda \to \infty$, yielding the second. Strict invertibility from Theorem 2.3 allows termwise inversion of each first-order expansion. Explicitly, $s = A\lambda + O(\lambda^2)$ implies $\lambda = s/A + O(s^2)$; and with $t = 1 - s$, $t = B/\lambda + O(\lambda^{-2})$ implies $1/\lambda = t/B + O(t^2)$ and hence $\lambda = B/t + O(1)$. □

4.1. Observed-difference plug-in mean

The previous results treat μ as fixed. The option

$$\widehat{\mu}(y) = \frac{1}{r}\mathbf{1}_r^\top D_d y =: g^\top y, \qquad g = D_d^\top \mathbf{1}_r/r, \tag{20}$$

instead defines a different *data-dependent linear mapping* $y \mapsto \widehat{\tau}_{s,\widehat{\mu}(y)}$ for every fixed target s. Its full Jacobian need not equal H_s.

Theorem 4.4 (Exact plug-in smoother and EDF correction)**.** *Let $s \in [0,1]$ be fixed and put $a = U^\top \mathbf{1}_r$, $b_s = V \operatorname{diag}(w(s))\Sigma^{-1}a$. The estimator obtained by substituting (20) into (11) is the linear map*

$$\widehat{\tau}_{s,\widehat{\mu}(y)} = T_s y, \qquad T_s = H_s + b_s g^\top, \tag{21}$$

where $H_s = I_n - V \operatorname{diag}(w(s))V^\top$. Its unconditional trace (holding s fixed, but not holding the estimated

mean fixed) is

$$\nu_{\mathrm{plug}}(s) = \operatorname{tr} T_s = n - rs + \Delta(s),$$
$$\Delta(s) = \frac{1}{r}\sum_{j=1}^{r} w_j(s) a_j^2, \qquad 0 \le \Delta(s) \le 1. \tag{22}$$

In particular,

$$\nu_{\mathrm{plug}}(0) = n, \qquad \nu_{\mathrm{plug}}(1) = d + 1,$$

whereas the conditional fixed-μ trace equals $n - rs$, with limiting value d at $s = 1$.

Proof. Equation (11) can be written $\widehat{\tau}_{s,\mu} = H_s y + \mu b_s$. Replacing μ by $g^\top y$ gives (21). Using $\operatorname{tr}(b_s g^\top) = g^\top b_s$ and $D_d = U\Sigma V^\top$,

$$\begin{aligned} g^\top b_s &= \frac{1}{r}\mathbf{1}_r^\top D_d V \operatorname{diag}(w)\Sigma^{-1} a \\ &= \frac{1}{r}\mathbf{1}_r^\top U\Sigma \operatorname{diag}(w)\Sigma^{-1}U^\top \mathbf{1}_r \\ &= \frac{1}{r}\, a^\top \operatorname{diag}(w) a = \frac{1}{r}\sum_j w_j a_j^2. \end{aligned}$$

Together with $\operatorname{tr} H_s = n - rs$ this proves (22). Since U is orthogonal, $\sum_j a_j^2 = \|\mathbf{1}_r\|_2^2 = r$. Every $w_j \in [0,1]$, so $\Delta \in [0,1]$. At $s = 0$ all weights are zero and $\Delta(0) = 0$; at $s = 1$ every weight is one, giving $\Delta(1) = 1$. The endpoint values follow. □

Corollary 4.5 (A caution for interpreting normalized smoothness)**.** *If μ is estimated from observed differences, the index s in (7) still specifies the* conditional *trace reduction $r^{-1}\sum_j w_j(s)$. The reduction calculated from the full plug-in operator is instead*

$$\frac{n - \operatorname{tr}(T_s)}{r} = s - \frac{\Delta(s)}{r}.$$

These two quantities differ for interior s, and their endpoint values at $s = 1$ are 1 and $1 - 1/r$, respectively.

Proof. Substitute (22). For $0 < s < 1$, the $w_j(s)$ are strictly positive, while $a \neq 0$ because U is orthogonal and $\mathbf{1}_r \neq 0$, so $\Delta(s) > 0$. The endpoint values follow from Theorem 4.4. □

Corollary 4.6 (Degenerate one-difference case)**.** *If $r = n - d = 1$ and $\widehat{\mu}(y) = D_d y$, then $\widehat{\tau}_{s,\widehat{\mu}(y)} = y$ for* every $s \in [0,1]$. *In particular, the full plug-in trace is n, even though the conditional trace equals $n - s$.*

Proof. For $\tau = y$ the residual in (2) is zero, and the penalty $D_d y - \widehat{\mu}(y)$ also vanishes. Thus $J_{\lambda,\widehat{\mu}(y)}(y) = 0$ for every finite $\lambda \ge 0$, the global minimum of the nonnegative objective. Uniqueness from Theorem 2.2 gives $\widehat{\tau} = y$. The limit $s = 1$ is also y by continuity; differentiating the identity map gives trace n. Alternatively, since there is only one positive spectral weight, Theorem 4.4 gives $\Delta(s) = w_1(s) = s$ and $\nu_{\mathrm{plug}}(s) = n - s + s = n$. □

Remark 4.7 (Degrees of freedom versus statistical claims)**.** The trace $\operatorname{tr}(T_s)$ is the exact divergence of the linear fitted-value map for a fixed s. Interpreting this divergence as a covariance-based effective number of parameters requires a statistical model (e.g., homoskedastic independent Gaussian errors). If s itself is selected using y, further data-dependence arises, and neither trace formula automatically describes the full adaptive procedure. Similarly, deleting one observation and holding a plug-in $\widehat{\mu}$ fixed is not an exact leave-one-out refit of the complete plug-in estimator.

5. Numerical algorithm, validation and cost

5.1. Root evaluation without an explicit penalty

Equation (9) provides a finite bracket for any interior s and a finite computed positive singular spectrum. For numerical work, form

$$\alpha_j = 2\log\sigma_j - \log\delta_*, \quad \log\delta_* = \operatorname{median}_j(2\log\sigma_j),$$

rather than computing δ_j and their ratios directly. Using the logistic function, evaluate $G(x) = r^{-1}\sum_j \operatorname{expit}(x + \alpha_j)$. When $s > 1/2$, it is preferable to solve the *complementary equation*

$$\frac{1}{r}\sum_{j=1}^{r} \operatorname{expit}(-x - \alpha_j) = 1 - s \tag{23}$$

rather than subtracting two nearby numbers after evaluating $G(x)$. Brent's bracketing method, bisection or safeguarded Newton iteration can find x_s; the strict monotonicity and bracket in Theorem 3.2 eliminate ambiguous roots. These choices do not imply a better condition number than log-λ inversion.

Direct-S reconstruction algorithm. Assume $D_d = U\Sigma V^\top$ has been computed and its positive singular values pass the chosen numerical-rank check.

1. Accept $s \in [0,1]$, $y \in \mathbb{R}^n$, and either a fixed finite μ or the explicit plug-in $\widehat{\mu}(y)$ of (20).
2. If $s = 0$, return $\widehat{\tau} = y$. If $s = 1$, set $w_j = 1$ for every positive singular direction. Otherwise compute the log-ratios α_j and bracket x using (9).
3. Solve $G(x) = s$ (or (23)) with a bracket-preserving scalar algorithm. Set $w_j = \operatorname{expit}(x + \alpha_j)$.
4. Form $c_\mu = V^\top y - \mu\Sigma^{-1}U^\top \mathbf{1}_r$ and return $\widehat{\tau} = y - V(w \odot c_\mu)$. Optionally return the *conditional* EDF $\nu_{\mathrm{cond}} = n - rs$, the attained mean $r^{-1}\sum_j w_j$ and a root residual. For a plug-in mean, separately report ν_{plug} from (22).

The first two endpoints use limiting weights, not matrix factorization at an infinite or zero penalty. The mathematical formula avoids forming the potentially large degree-d particular polynomial p_μ in the interior.

Corollary 5.1 (Residual-to-trend error certificate)**.** *Suppose an approximate scalar solution* $\widetilde{x} \in \mathbb{R}$ *yields computed* exact-arithmetic *weights* $\widetilde{w}_j = \operatorname{expit}(\widetilde{x} + \alpha_j)$ *with scalar residual* $\varepsilon = G(\widetilde{x}) - s$*. Let* $\widetilde{\tau} = y - V(\widetilde{w} \odot c_\mu)$*. Then*

$$\|\widetilde{\tau} - \widehat{\tau}_{s,\mu}\|_2 \le r|\varepsilon|\,\|c_\mu\|_2 .$$

Proof. Since G is strictly increasing and maps $\mathbb{R}$ onto $(0,1)$, $\widetilde{w} = w(\widetilde{s})$ with $\widetilde{s} = G(\widetilde{x}) = s + \varepsilon$. Thus $\widetilde{\tau} = \widehat{\tau}_{\widetilde{s},\mu}$; apply (15). □

Remark 5.2 (Scope of the certificate)**.** This is an *algebraic root-residual* bound. A floating-point implementation also incurs errors in its computed SVD, scalar logistic sums and matrix products. The certificate alone does not bound those errors, and it cannot establish that the new approach is more accurate than computing the same weights through a carefully stabilized log-λ algorithm.

5.2. Arithmetic, storage and reuse

Let I denote the number of scalar residual evaluations. For dense reduced SVD of an $r \times n$ matrix with $r \le n$, standard algorithms take $O(nr^2)$ setup work and $O(nr)$ storage. Solving the scalar equation requires $O(Ir)$ work; reconstructing one full n-vector after the SVD costs $O(nr)$ for spectral projections and multiplication. Calculating $\widehat{\mu}(y) = r^{-1}\mathbf{1}_r^\top D_d y$ requires up to $O(nd)$ arithmetic using a difference stencil. No dense $n \times n$ inverse is formed.

If K fixed smoothness targets are used on R series of the same length and difference order, the weights can in principle be precomputed once and stored with $O(Kr)$ additional memory. The total major operations are

$$O\big(nr^2 + KIr + RKnr\big) \tag{24}$$

for RK complete reconstructions after shared preparation; series-dependent projections may also be shared. For only a single fit, structured banded factorization may be asymptotically cheaper and should be treated as an essential competitor (Weinert, 2007). A cached inverse-penalty implementation could cache *the identical weights*; thus (24) is a description of the proposed execution architecture, not a proof of superiority over a well-engineered baseline.

5.3. What can and cannot be verified from the current repository

The implementation `src/trendestimation/smoothness_weights.py` already evaluates direct log-odds weights and reconstructs the trend without forming λ. Its tests compare results against the existing `lambda_for_smoothness()+fit()` path for difference orders 1 through 6, fixed nonzero μ, the observed-difference plug-in and endpoint cases. These tests support finite-precision agreement on their sampled cases, not a globally valid numerical error bound. No new numerical experiments are performed in this manuscript, and no numerical tables in the preceding two CSDA drafts are reclassified as a direct-weight runtime comparison.

6. Discussion, falsifiable claims and research agenda

6.1. What has been proved

The PLS objective (2) is unchanged. Theorem 3.1 characterizes exactly which shrinkage weights belong to a single quadratic-penalty family; Theorem 3.2 proves a unique scalar root for every prescribed interior s and provides a finite bracket; Theorem 3.4 proves exact equivalence between penalty-based and direct-weight fits, including fixed nonzero means of differences and both endpoints. Theorem 4.1 quantifies sensitivity and separates weight stability from scalar-root conditioning. Theorem 4.4 supplies the full Jacobian trace when the difference mean is computed from the observations. The results are algebraic consequences of an established linear smoother. Their formal presentation is not evidence that every identity is historically novel.

6.2. Why using weights can nevertheless be useful

For an interior target s and a known positive spectrum, the weight representation computes all shrinkage factors from a finite scalar x. In exact arithmetic, $\lambda = \exp(x)/\delta_*$ from (10); the direct-weight fitting path has no need to evaluate that expression. It may therefore be possible to reconstruct a trend in a regime where λ itself cannot be represented as a finite floating-point number, provided the singular values and the weight root remain numerically resolvable. This is a *conditional representation advantage*, not a universal proof of greater numerical accuracy. If the smallest singular values have already been corrupted or lost to underflow, the method cannot recover that information.

Caching $w(s)$ for repeated series with identical (n, d, s) also avoids repeated inversion. An equally optimized penalty-based implementation can cache the same weights, so no intrinsic asymptotic advantage follows from the new interface. The correct comparison distinguishes (i) initial SVD cost, (ii) scalar inversion, (iii) repeated full-vector reconstruction, (iv) memory, and (v) failed or inaccurate candidate fits. Prior structured Whittaker–Henderson algorithms must remain in the comparison (Weinert, 2007; Biessy, 2026).

6.3. Proposed Experiment006: a falsifiable validation

A later *separately numbered* experiment should compare three implementations on identical inputs: (A) stable log-λ inversion followed by shift-free spectral fitting; (B) the present direct log-odds weight fit; and (C) the direct-weight path with cached weights and

Table 1
Mathematical and computational status of three equivalent implementations. The final comparison must time each phase at identical s, d, μ, precision and hardware, rather than infer a speedup from eliminating the name of λ.

Execution path	Scalar operation	Reconstructing the trend	Current evidence
Explicit $\Lambda_d(s)$ plus spectral PLS	Invert $S_d(\lambda)$, retain λ	Evaluate $w_j(\lambda)$ and shift-free SVD	Existing validated baseline
Implicit direct weights	Solve $G(x) = s$, do not form λ	Evaluate $w_j = \mathrm{expit}(x + \alpha_j)$ and shift-free SVD	Implemented; algebraically equivalent
Cached weights across series	Solve one scalar problem per distinct (n, d, s)	Reuse w for each new y, μ	Proposed timed comparison

repeated observations. Use $d = 1, \ldots, 6$, multiple n values, fixed $\mu = 0$ and fixed nonzero μ, as well as the separately labeled observed-difference plug-in. Cover an interior s grid and attainable values near zero and one.

At every identical (n, d, s, μ, y), measure total and phase-separated time, root evaluations, allocation and peak memory, attained smoothness, method failure, numerical relative error against an independent high-precision *same-estimator* reference, and statistical error against known latent trends when synthetic truth is available. Include the same scalar precision tolerance and a cache-equivalent penalty-based baseline. Block independent replicates by simulated series and machine run; do not treat hundreds of s-grid points as independent samples. Failure counts should remain in the raw data. A direct-weight advantage would be supported by matched measurements and high-precision accuracy at the same requested s, not by an isolated microbenchmark or a faster result from a different candidate grid.

Two especially testable predictions are: (i) a direct-weight path may avoid forming an overflowing λ in some high-order, high-smoothness cases where the log-singular spectrum remains usable; and (ii) caching weights will be most beneficial when one (n, d, s) configuration is reused across many observations. Neither prediction is claimed as verified here. An equally optimized log-λ and weight-cache baseline might match both outcomes.

6.4. Limits and scope for a standalone paper

Our results concern unweighted, equally indexed finite differences and quadratic penalties. They do not establish a theory of irregular observation spacings, time-varying smoothness, ℓ_1 trend filtering, generalized likelihoods, or a new forecast-selection criterion. The smoothness coordinate is a normalized *conditional trace index*; when μ is estimated from data the full fitted-value Jacobian has the additional term in (22). If s is selected from the data, its adaptive effect requires further analysis. A bounded parameter interval does not imply a globally well-conditioned inverse or a globally optimal finite search.

7. Conclusion

A PLS trend can be reconstructed by solving one monotone scalar equation for a coupled vector of spectral weights and applying a shift-free SVD formula, without explicitly calculating the penalty coefficient in the fitted-trend path. The estimator, its shrinkage family, and the implicit scalar inversion remain those of the classical penalized problem. The exact endpoint projection, sensitivity bound and plug-in degrees-of-freedom correction help clarify what prescribing $s \in [0, 1]$ mathematically controls. Whether this direct execution path is preferable in practice is an open empirical question for the proposed paired precision–runtime benchmark.

Software and evidence status

The current implementation is maintained in https://github.com/heri-espino/trendestimation, module `src/trendestimation/smoothness_weights.py`; existing equivalence tests reside in `tests/test_spectral_smoothn`. This document is a theory-first research draft. Its theorems have proofs, but no new runtime or statistical Monte Carlo measurements were conducted for the manuscript. Authorship, journal target and statistical claims are provisional pending independent coauthor review.

References

, 2026. R: Fit a Smoothing Spline, stats::smooth.spline. R Core Team. URL: https://stat.ethz.ch/R-manual/R-devel/library/stats/html/smooth.spline.html. documentation of prescribed effective degrees of freedom; accessed 2026-11-01.

Biessy, G., 2026. Whittaker–henderson smoothing revisited: A modern statistical framework for practical use. ASTIN Bulletin 56, 1–31. doi:10.1017/asb.2025.10061.

Cortés-Toto, D., Guerrero, V.M., Reyes, H.J., 2017. Trend smoothness achieved by penalized least squares with the smoothing parameter chosen by optimality criteria. Communications in Statistics—Simulation and Computation 46, 1492–1507. doi:10.1080/03610918.2015.1005236.

Guerrero, V.M., 2007. Time series smoothing by penalized least squares. Statistics & Probability Letters 77, 1225–1234. doi:10.1016/j.spl.2007.03.006.

Guerrero, V.M., 2008. Estimating trends with percentage of smoothness chosen by the user. International Statistical

Review 76, 187–202. doi:10.1111/j.1751-5823.2008.00047.x.

Weinert, H.L., 2007. Efficient computation for whittaker–henderson smoothing. Computational Statistics & Data Analysis 52, 959–974. doi:10.1016/j.csda.2006.11.038.

Whittaker, E.T., 1923. On a new method of graduation. Proceedings of the Edinburgh Mathematical Society 41, 63–75. doi:10.1017/S0013091500077853.